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**TITLE: DESIGN AND FABRICATION OF VERTICAL AXIS WIND
TURBINE
BY**

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The report is submitted in partial fulfillment of requirements for the award of degree bachelor of Engineering in mechanical and production engineering

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DEDICATION

We hereby dedicate this work, to work to the Almighty God, who has brought us this far, our families, lecturers and friends for their kindness, devotion and endless support and efforts in guidance and finance.

DECLARATION

We declare that this is our original work and it has not been submitted elsewhere for marking for achievement of any academic award. It is a result of our research except otherwise stated. All sources used in production of the report are acknowledged by citation and references.

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DECLARATION BY THE SUPERVISOR

This is to certify that the project titled “DESIGN AND FABRICATION OF A VERTICAL AXIS WIND TURBINE” submitted by the students whose names appear above is an authentic work done by them under my supervision.

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Signature.....

EXECUTIVE SUMMARY

Kenya faces problems in increasing the share of wind energy in the state's electricity mix. It is difficult to add more Horizontal Axis Wind Turbines (HAWTs) to wind farms because of the negative impacts they create for each other when placed too close together. It is also increasingly expensive to permit, buy land, build roads and provide transmission lines for new wind farms. Further, large HAWTs pose threats to migratory and native bird populations, resulting in additional costs and difficulties in obtaining permits and developing environmental impact mitigation plans.

The use of Vertical Axis Wind Turbines (VAWTs) as a solution to these problems has not yet been investigated, due to lack utility scale VAWTS and data on their impacts to neighboring HAWTs and wildlife. Before wind farm owners will allow the large-scale deployment of VAWTs near HAWTs, field research must demonstrate that wakes produced by VAWTs have neutral or positive effects on the energy production and maintenance of nearby HAWTs. Before permits can be obtained for installation of VAWTs in most Kenya wind farms, research must demonstrate that the turbines do not negatively impact bird populations.

The technical and economic feasibility of Vertical Axis Wind Turbines [VAWTs] has also been largely unexplored to date, but the inherent advantages of VAWT suggests a transformational opportunity to allow access to vast wind resource sites such as off the East coast of Kenya and north part of Kenya.

In this project, we will:

- Design a Vertical axis Wind Turbine
- Check efficiency of the VAWT
- Provide suggestion of if it can be implemented

TABLE OF CONTENTS

ACKNOWLEDGEMENT	i
DEDICATION	ii
DECLARATION.....	iii
EXECUTIVE SUMMARY	iv
Chapter 1.....	1
INTRODUCTION	1
1.1 PROJECT DEFINITION	1
1.2 PROJECT OBJECTIVES.....	1
Chapter 2.....	2
LITERATURE REVIEW	2
2.1 PROJECT BACKGROUND	2
2.2 PREVIOUS WORK.....	5
2.2 Working principle of VAWTs.....	6
2.3 Wind Energy.....	7
2.3.2 Wind Speed.....	7
2.4 Wind Turbine Classification	9
2.5 Savonius VAWT	11
2.6 Background to WASP.....	12
2.8 Mounting system	12
2.9 Noise and Vibrations	13
CHAPTER 3.....	14
METHODOLOGY	14
3.1 INTRODUCTION	14
3.2 MAJOR COMPONENTS OF VAWT	16
FABRICATION PROCESS	16
3.3 Fabrication of central Shaft.....	16
3.4 Adjusting the Tool Bit	17
3.5 Preparing to Drill	20
3.6 Fabrication of washers.....	21
3.7 Fabrication of turbine blade	23
3.8 Fabricating Lower Support Shaft.....	24
3.9 Modification of Hub.....	25

3.10	Modification of Bearings.....	25
3.11	Hub shell and flanges.....	26
3.12	Rim.....	26
3.13	Spokes.....	27
	Connecting the Alternator	28
3.14	How the alternator works.....	30
3.15	Regulating the current to the battery	31
	BATTERY	31
3.16	Car battery function: Chemical energy becomes electrical energy.....	32
	FINAL APPEARANCE OF OUR FABRICATED VAWT	35
	CHAPTER 4.....	36
	RESULTS.....	36
4.1	THEORY	36
4.2	MATHEMATICAL MODEL.....	36
	=1.3KW	40
	ACTUAL/ observed VALUES	40
	EFFICIENCY	43
	CHAPTER 5.....	45
	CONCLUSION AND RECOMMENDATION	45
	RECOMMENDATION	45
	APPENDICES	47
	REFERENCES.	50

List of figures

Figure 2.1: horizontal axis wind turbine.....	3
Figure 2.2: vertical axis wind turbine.....	4
Figure 2.3: comparison of HAWT and VAWT.....	4
Figure 2.4: correlation of height with wind speed and power.....	8
Figure 2.5: HAWT.....	10
Figure 2.6: aero-foil wing.....	10
Figure 2.7: VAWT.....	11
Figure 2.8: savanious VAWT.....	12
Figure 3.1: model (a).....	14
Figure 3.2: model (b).....	15
Figure 3.3: turning process.....	17
Figure 3.4: machined shaft.....	18
Figure 3.5: Facing process.....	18
Figure 3.6: Various sizes of centre drills.....	19
Figure 3.7: Drilling processes.....	20
Figure 3.8: Partly drilled hole necessary before turning operations.....	20
Figure 3.9: Our team performing drilling operations.....	21
Figure 3.10: washer.....	21
Figure 3.11: Thin plates for making a washer.....	22
Figure 3.12: Diagonals on the two plates to identify centre.....	22
Fig 3.13: Making a washer with lathe machine.....	23
Figure 3.14: Scoops.....	24
Figure 3.15: Lower Support Shaft.....	24
Figure 3.16: Boring tool.....	25
Figure 3.17: Bearing.....	26
Figure 3.18: Hub Shell.....	26
Figure 3.19: RIM.....	26
Figure 3.20: kidride hub.....	26
Figure 3.21: alternator.....	27
Figure 3.22: Bottom of alternator.....	28

Fig 3.23: Inside an alternator.....	29
Figure 3.24: Battery.....	33
Figure 3.25: final part.....	35
Figure 4.1: Block diagram representing our VAWT charging system.....	41
Figure 4.2: alternator and turbine speeds.....	42
APPENDIX 1.....	46
APPENDIX 2.....	47
APPENDIX 3.....	47
APPENDIX 4.....	48

List of tables

Table 4.1: Speed difference between engine and alternator.....	42
Table 4.2: battery voltage.....	43
Table 4.3: power produced.....	43

CHAPTER 1

INTRODUCTION

1.1 PROJECT DEFINITION

This project is about designing and manufacturing of a Wind Turbine that can convert wind by using Vertical Axis Wind Turbines (VAWT) to a useful energy. The current power demand in Kenya is very high compared to power consumption average. This high demanding should take the focus of attention in thinking in different sources of energy.

One of the best sources of energy that can apply the concept of sustainability is renewable energy such as sun, wind, and rivers. The positive point of wind energy is that unlike solar energy that only used with sunlight, wind turbine can be useful all the 24 hours all the year.

Another concept of sustainability is the way that we should use in utilizing this renewable energy efficiently, and environmentally friendly. This in turn will eliminate the environment hazard and improve Kenya's health and life style.

Streets, public parks, schools and public facilities are consider as main power consumers, these consumers should be vulnerable to wind from time to time. The idea of this project is to convert this wind by using Vertical Axis Wind Turbines (VAWT) to a useful energy by using it as a power source that can serve these consumers.

1.2 PROJECT OBJECTIVES

Main objective

The main objective of the project is to **design and fabricate a small scale vertical axis wind turbine.**

Specific objectives

1. To collect wind data around Moi University and analyze it.
2. To survey and identify the best location for its installation.
3. To identify best materials for use in fabrication.
4. To recommend if its use is efficient

CHAPTER 2

LITERATURE REVIEW

2.1 PROJECT BACKGROUND

Energy is the main economy base of any country. Sources of energy are not easy to have. Having multiple sources of energy is extremely important to secure the basic living requirement of any country. Utilizing the nature could help in converting some of the natural phenomena such as sun, wind, sea and oil into useful energy. This kind of energy called renewable energy. Science Daily Research Newspaper has defined renewable energy as a form of energy resource that is replaced rapidly by a natural process such as power generated from the sun or from the wind. Recently, the increasing demand of renewable energy is very well noticed. According to a report by the International Energy Agency, the increase of amount of electricity produced from renewable sources increased from just over 13% in 2012 to 22% the following year.

They also predict that that figure should hit 26% by 2020.

The traditional power plants in Kenya are mainly working on the fuel either gas or oil which are not environmental friendly.

EcoSpark environmental charity has considered oil power plants as one of the most contributors of environment pollution. EcoSpark environmental charity has listed the below most significant environmental impacts:

- Oil causes air pollution and greenhouse gas emissions.
- Oil uses large amounts of water, and creates water pollution and thermal discharge.
- Oil creates hazardous sludge and solid waste.
- Extracting and refining oil is environmentally destructive.
- Transporting oil is risky and can harm the environment.
- Oil is a non-renewable electricity source.

Such of the above environment affects lead us to think seriously about the renewable energy sources, which will eliminate the environment hazard and improve health and life style.

Wind energy is one of the most important energy sources. The concept of wind energy is transforming the wind's kinetic energy into mechanical energy. This energy drive blades that turn generators that produce electricity. Our project is fitting with wind energy source. The idea of this project is to convert the wind by using Vertical Axis Wind Turbines (VAWT) into power.

They are two types of wind turbines, **Horizontal Axis Wind Turbines (HAWT)** as shown in figure 2.1 that is more commonly used across the world and they are used as a power plants.

These kind of turbines are the most efficient of wind turbine.

However, the blade span of horizontal wind turbines is larger than vertical axis machines which limits placement confined spaces. Some people also find the large blade area of horizontal axis machines objectionable.

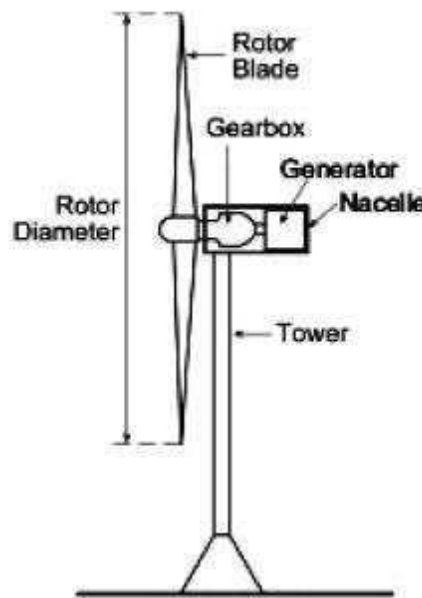


Figure 2.1: Horizontal Axis Wind Turbine

The other type of wind turbine is the **Vertical Axis Wind Turbines (VAWT)** as shown in figure 2.2. VAWT is the most popular of the turbines that people are using to make their home a source of renewable energy.

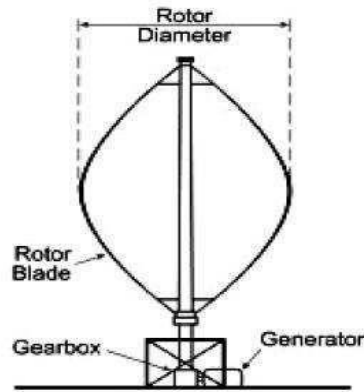


Figure 2.2: Vertical Axis Wind Turbine

VAWT is not as commonly used as the Horizontal Axis Wind Turbine. The reason behind that is that VAWT is less efficient than HAWT when considered as a power plant generator. However, for the small scales like homes, parks, or offices VAWT is more efficient.

Vertical axis turbines are powered by wind coming from all 360 degrees, and even some turbines are powered when the wind blows from top to bottom. Because of this versatility, vertical axis wind turbines are thought to be ideal for installations where wind conditions are not consistent, or due to public ordinances the turbine cannot be placed high enough to benefit from steady wind. Figure 3 shows the configuration of HAWT vs VAWT.

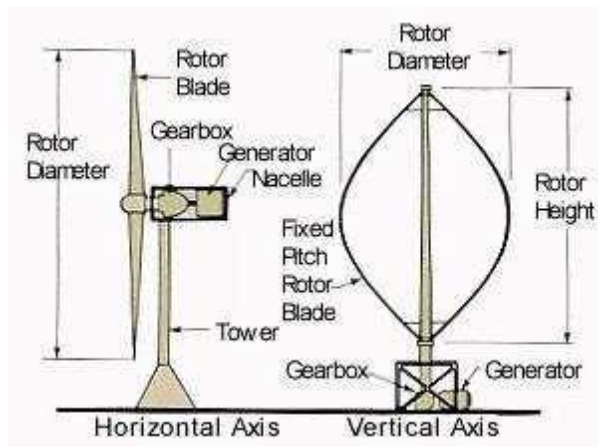


Figure 2.3: Comparison of HAWT and VAWT

2.2 PREVIOUS WORK

There are two different styles of vertical wind turbines. One is the Savonius model, which is our project is based on, and the other type is the Darrieus model. The first model looks like a gallon drum that is been cut in half with the halves placed onto a rotating shaft. The second model is smaller and looks much like an egg beater. Most of the wind turbines being used today are the Savonius models.

Renewable Energy UK website provided some information about these two model.

“A Savonius is a type of vertical axis wind turbine (VAWT) generator invented in 1922 by Sigurd Johannes Savonius from Finland though similar wind turbine designs had been attempted in previous centuries.”

“A Darrieus is a type of vertical axis wind turbine (VAWT) generator. Unlike the Savonius wind turbine, the Darrieus is a lift-type VAWT. Rather than collecting the wind in cups dragging the turbine around, a Darrieus uses lift forces generated by the wind hitting aerofoils to create rotation.”

In Jun 2.15, International Research Journal of Engineering and Technology (IRJET) has Published a research titled “DESIGN, ANALYSIS AND FABRICATION OF SAVONIUS VERTICAL AXIS WIND TURBINE”

This research discussion was to showcase the efficiency of Savonius model in varying wind conditions as compared to the traditional horizontal axis wind turbine. It evaluated some observation that showed that at low angles of attack the lift force also contributes to the overall torque generation. Thus, it can be concluded that the Savonius rotor is not a solely drag-driven machine but a combination of a drag-driven and lift-driven device.

Therefore, it can go beyond the limit of Maximum power coefficient C_p established for the purely drag-driven machines.

Some of this researched conclusions are that the vertical axis wind turbine is a small power generating unit with the help of free source of wind energy. It is designed under consideration of household use. Generally, At least 10% power of the consumption can be fulfil by the Savonius model.

The research has also resulted that this turbine is generally suitable for 8 to 10m of height above ground level. Because at ground level velocity of air is very less. And finally the alternate option for turbine blade material is reinforced glass fiber because of its more elastic nature but it is costlier than aluminum alloy. To have the best efficiency of the power output from our turbine, we have done some brainstorming in what are the most significant factor that affect the turbine, the blade angle was agreed to be the most significant one.

By doing some researches, we found an article that focusing in the turbine blade angle. A research article published by Advances in Mechanical Engineering (AIME) with a title of “EFFECT OF THE BLADE ARC ANGLE ON THE PERFORMANCE OF A SEVONIUS WIND TURBINE” [10].

2.2 WORKING PRINCIPLE OF VAWTS

a) Savonius Wind Turbines: The Resistance Runner

Savonius wind turbines have blades built around the vertical shaft in a helix form, which basically looks like DNA or fusilli pasta. Wide, solid wind-receiving area of the blades is one of the most significant features of a Savonius wind turbine.

When in operation, Savonius wind turbines rely on the flow resistance mechanism to turn their rotors. In simple words, the dynamic pressure of the wind against the blades pushes the rotor into rotation. At the same time, the opposite side of the blades encounters a force of aerodynamic resistance or “drag.” This is just like what we experience when cycling or running: There’s always the air flow coming against us. Because of this, Savonius wind turbines can only turn as fast as the wind speed.

b) Darrieus: The Uplift Runner

Often called the “egg-beater” and named after the French inventor Georges Darrieus, classic Darrieus turbines have long, curved wings with each end attached to the top and bottom of the rotor shaft. Another model of the Darrieus turbines has three straight wings connected to the shaft parallelly, forming the “H” shape.

In terms of operation, Darrieus utilizes the “lift” aerodynamic force to rotate. By flowing around the structure, the wind creates a suction on the front side of the turbine, driving the wings to rotate. Because of the shape of the wings, they do not experience as much drag as Savonius turbines do. Once the rotation starts, Darrieus wind turbines are able to accelerate to rotate faster than the wind speed.

2.3 WIND ENERGY

Wind is generated from solar energy unevenly heating the earth. This uneven heating creates pressure changes in the atmosphere, generating wind. This wind can then be harnessed by a wind turbine. As the wind pushes the blades of a turbine, a generator attached to the axis of the shaft and when spun creates electricity that can be sent to the grid and used in households for electricity. (windies.gov, 2012)

Wind turbines are a clean way to generate power, yet there are many significant problems with them as well. One problem is that they are extremely expensive to design and install, and in order to generate enough energy for communities and cities require space for wind farms. Another issue is that they have to be created in locations where there is enough wind energy to generate enough electricity to justify the cost of the machine.

2.3.1 Power Density

Geography can greatly effect wind speed, and in effect the power from the wind. Knowing this information prior to setting up a wind turbine is imperative. Calculating the average power from wind is a simple equation:

$$P = \frac{1}{2} \rho v^3 A \dots\dots\dots (!)$$

Equation 1 indicates the importance of wind speed in power generation because power generation increases proportionally as wind increases to the third power. Knowing the power density will allow wind turbines to be placed in efficient locations for generating electricity.

$$V_{(z)} = V_{(z_{ref})} \left[\frac{z}{z_{ref}} \right]^\alpha \dots\dots\dots (2)$$

2.3.2 WIND SPEED

Another important factor is the height of the turbine rotor. One of the major reasons wind turbine costs are so high is because the higher altitude the turbine is located, the higher the velocity of the wind, which in turn increases the power output from the generator. Equation 2 is the power model which estimates the effect that height has on wind.

$V(z_{ref})$ is the reference point that can be looked up in a chart, z is the height above the ground, and α is the power law exponent in which is affected by the surface geometry of the land and needs to be researched. Figure 4 contains equation 2 against height to show how wind velocities exponentially affect power output.

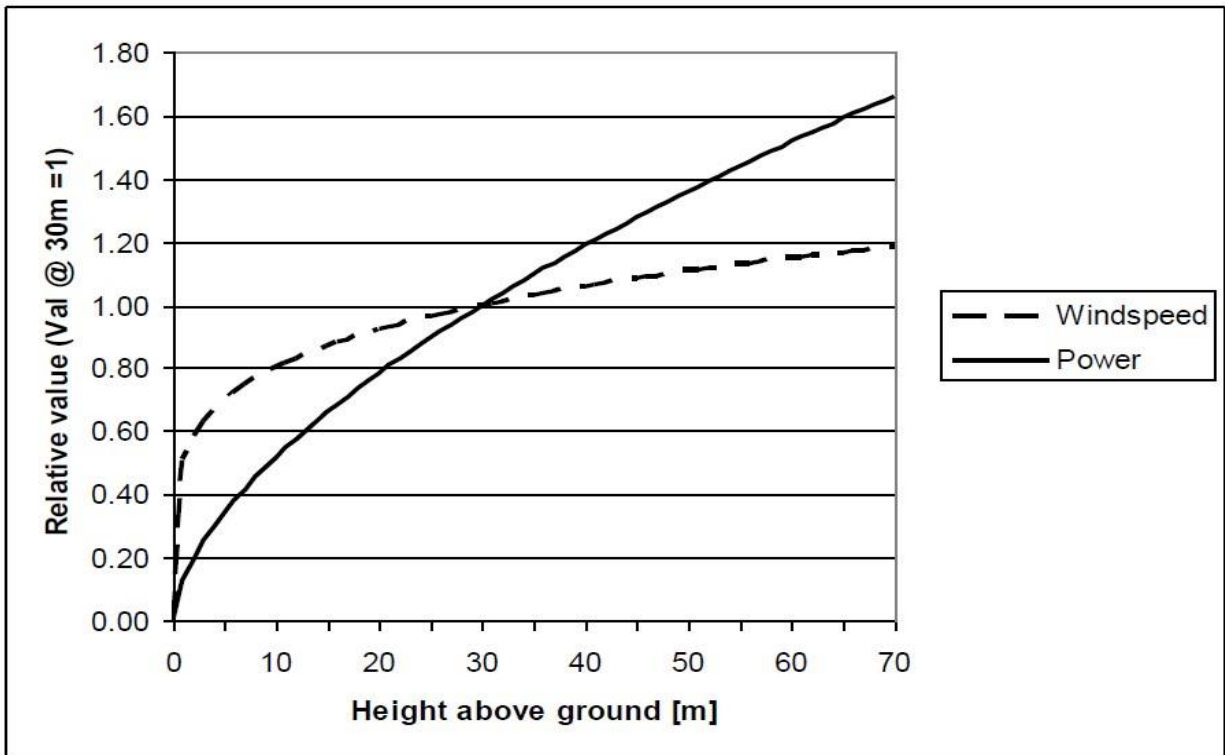


Figure 2.4: Correlation of height with wind speed and power

As figure 4 indicates, below 30m in the air the wind velocity increases at a faster rate than the corresponding power density. However, once an altitude of 30m is reached, the power density increases at a faster rate than wind velocities increases. This demonstrates that the higher a wind turbine is, the more energy can be obtained from the turbine.

2.2.3 Power Coefficient

The power coefficient is the percentage of power received by the wind turbine through the swept area of the turbine blades. Equation depicts how to calculate the coefficient of power.

$$C_p = 4 \frac{V_t^2}{V_u^2} \left[1 - \frac{V_t}{V_u} \right] \dots \dots \dots (3)$$

In equation 3, V_u is the velocity of the wind as it approaches the wind turbine and V_t is the velocity of the wind as it passes through the swept area of the wind turbine blades. The maximum theoretically possible coefficient of power is called the Betz limit which is 0.593. Most current turbines today have a power coefficient between 0.3 and 0.4.

$$\alpha = \frac{\pi R}{V_u} \dots \dots \dots (4)$$

2.3.4 Tip Speed Ratio

Equation 4 defines the tip speed ratio is the ratio of the tip speed of the blade divided by the wind speed. The equation for tip speed ratio is described below:

In equation 4, λ is the tip speed ratio, Ω is the rotor rotational speed in radians per second, R is the rotor radius in meters and V_u is the wind speed.

2.4 WIND TURBINE CLASSIFICATION

The two major classifications of wind turbines are horizontal and vertical axis wind turbines, (HAWT and VAWT). The horizontal axis wind turbines are the most common and have blades rotate on an axis parallel to the ground.

2.4.1 HAWT

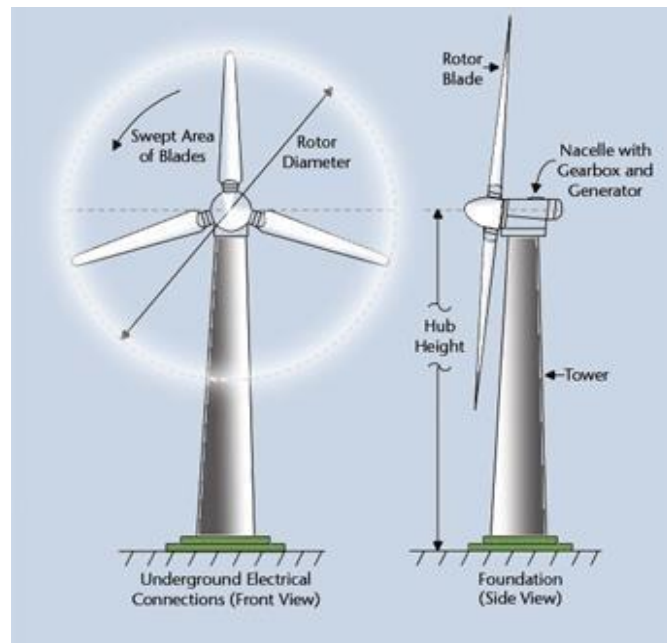


Figure 2.5: HAWT

Figure 5 shows a typical Horizontal axis wind turbines. The horizontal axis axil is attached by bearings at the top of the tower were the blades are also attached to the axel.

The axil is enclosed in a nacelle. The nacelle is where the gearbox and generator are located. Horizontal axis wind turbines utilize airfoil design to generate the spinning of the blades. The concept of the wind foil of a HAWT blade is that the wind travels over the top of the blade rather than under it, creating less pressure on top of the blade generating lift and creating rotational movement. Figure 6 depicts this information.

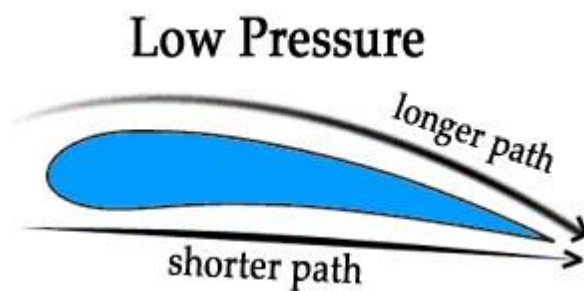


Figure 2.6: Aerofoil wing

2.4.2 VAWT

The other major classification for wind turbines are vertical axis wind turbines. These turbines spin on a vertical axis. Figure 2.8 is an example of a Darrieus vertical axis wind turbine. This turbine is an example of a commercially used vertical axis turbine. One of the major problems with vertical axis wind turbines is that an initial force is required to start the turbine's spinning. Another issue is that they are difficult to be designed for high altitudes. The blades on a vertical axis wind turbine can utilize an airfoil design like the VAWT; however a VAWT can also use blades that directly face the wind, as shown in figure 8.



Figure 2.7: VAWT

2.4.5 The Effects of Shrouds on Vertical Axis Wind Turbines (VAWT)

Research confirms that a three bladed turbine design with air foil blades outperform and is more efficient at lower speeds then the equivalent flat blade design.

2.5 SAVONIUS VAWT

The design utilizes an open overlapping two half cup designs that is very beneficial to wind turbine design. Some of the most appealing benefits of the Savonius design are it simple and cheap to construct, it has low noise and angular velocity when in use, and it can accept wind from any direction and can withstand extreme weather conditions without significant damage. In addition there are multiple variations to the design that change the performance of the turbine depending on blade configuration.

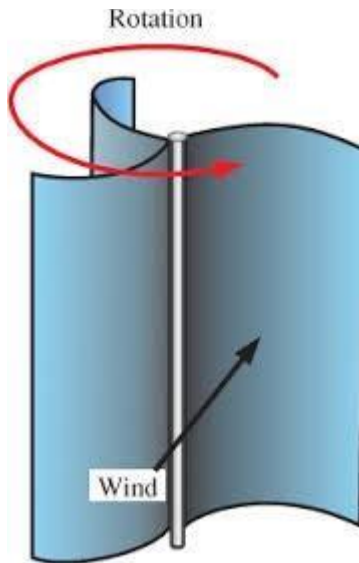


Figure 2.8: savanious VAWT

2.6 BACKGROUND TO WASP

WASP is a PC-program for the vertical and horizontal calculation of wind climate statistics. It contains different models to describe the wind flow over different areas and terrains.

This program is mainly for predicting wind climates, wind resources, and power productions from wind turbines and wind farms. The predictions are based on wind data measured at meteorological stations in the same region. The program includes a complex terrain flow model, a roughness change model, a model for sheltering obstacles, and a wake model.

2.8 MOUNTING SYSTEM

The most common method for a mounting structure for wind turbines is a monopole design. This consists of some sort of base, usually concrete, with a steel structured pole that extends to the owners desired height. As the progression of turbines has grown there has been desire for roof mounted systems. These roof mounted systems have not had the amount of research as the traditional monopole design. As the growth of roof mounted turbines rises there is an urge to design out the flaws that has been shadowing previous roof mounted systems.

2.9 NOISE AND VIBRATIONS

Wind turbines can create a constant humming noise that is considered an annoyance as well as produce vibrations that over time can ruin the integrity of a roof. This has hindered the popularity of consumers wanting to have wind turbines mounted to their roof. As a wind turbine is spinning and producing electricity it creates a constant vibration. This constant vibration can damage the shingles around the base as well as damage the trusses around the area where they are mounted. These vibrations are very difficult to prevent, so it is important to have a mounting system that will disperse the vibrations before reaching the actual structure of the house. This will dissuade a consumer away from having a wind turbine mounted on the roof.

Another issue that dissuades a consumer from using wind turbines is the constant humming noise produced when a turbine is generating electricity. A VAWT as opposed to a horizontal axis turbine does not produce as much noise as the traditional turbines due to design differences of the turbines as well as the path of motion of the blade. Another reason that VAWT are more for residential areas is their ability to operate at peak efficiency with turbulence that is produced by roof contours. When wind is traveling over roof peaks and different slopes of roofs they produce a turbulent wind pattern that actually disrupts horizontal axis turbines from producing electricity. This turbulent wind pattern caused by roof peaks does not disrupt the functioning of VAWTs, as well as produce as loud of noise compared to HAWT systems.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

Prior any appropriate solution can be developed, a thorough investigation has to be conducted in order to find out what solutions have already been proposed (information gathering). Once these solutions have been analyzed and the team has an understanding of why the respective solutions are not currently being implemented, a solution generation phase is taking place. Here various solutions are presented and evaluated against criteria and constraints

(concept generation). Solution concepts are then modeling

The results of the models are then analyzed and the model, as well as solution parameters, may be tweaked (model analysis and refinement).

Once the team has satisfactorily modeled all solution concepts of interest, the concept that performs best analytically, in addition to meeting all criteria and constraints, is selected (concept selection). The analytical model may then be verified experimentally, using a small scale modeling scheme or through a full scale experimental model.

We designed two models as shown



Fig 3.1: Model (a)

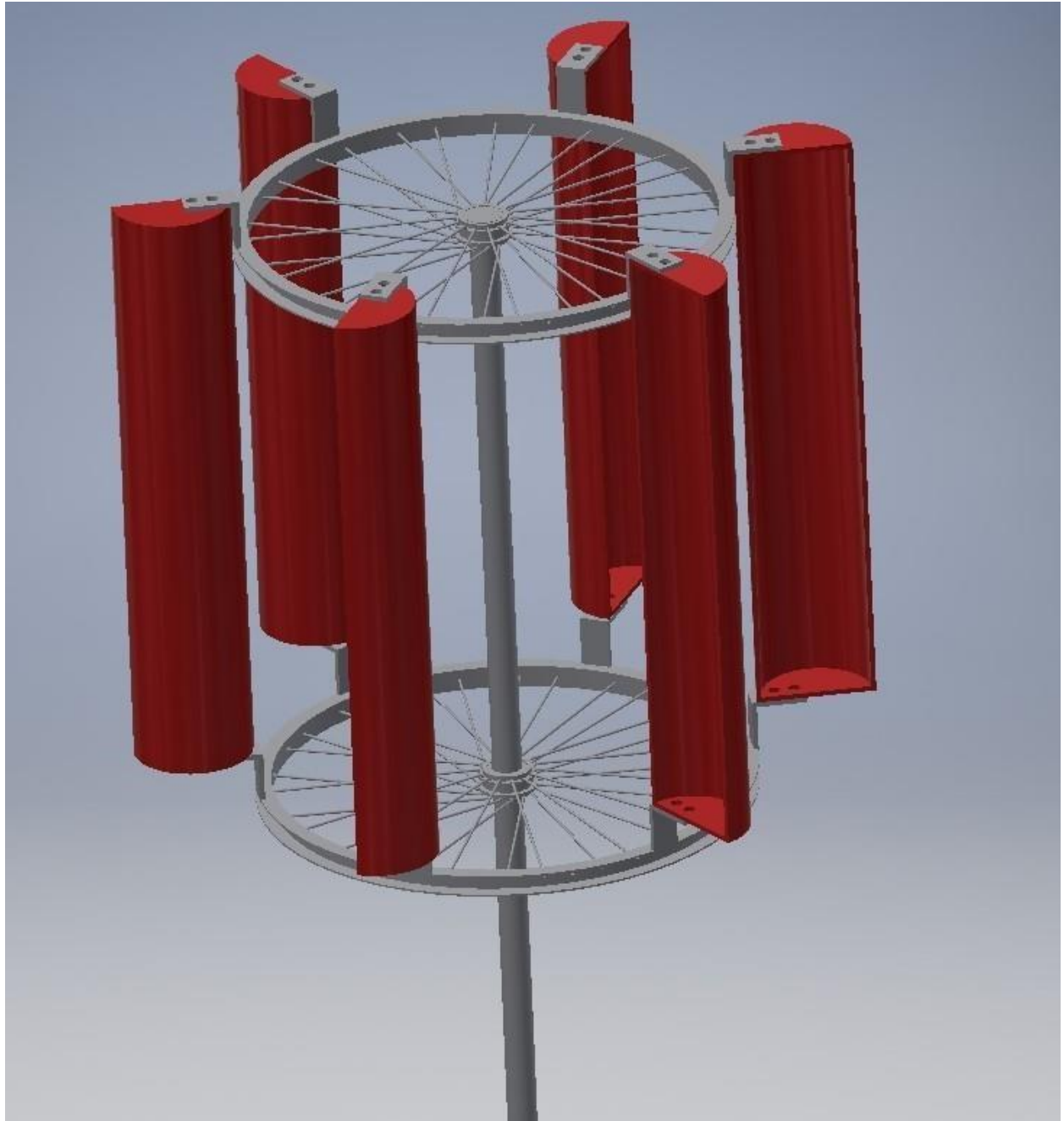


Figure 3.2: Model (b)

Through SOLID WORKS software, model (b) was much efficient. Therefore we concluded to fabricate model (b).

3.2 MAJOR COMPONENTS OF VAWT

- | | |
|------------------|------------|
| 1. Turbine blade | 6. Axle |
| 2. Shaft | 7. Rim |
| 3. Alternator | 8. battery |
| 4. Hub | 9. Belt |
| 5. Spokes | |

FABRICATION PROCESS

3.3 FABRICATION OF CENTRAL SHAFT

The shaft is the part that supports the two rims and base. We used a hollow steel metal of 17mm diameter.

Activities done to the shaft

1. Facing
2. Turning
3. Drilling

Turning

Turning is the removal of metal from the outer diameter of a rotating cylindrical workpiece. Turning is used to reduce the diameter of the workpiece, usually to a specified dimension, and to produce a smooth finish on the metal. Often the workpiece will be turned so that adjacent sections have different diameters.

Cutting Tool: *Turning Tool*

✓ a *depth of cut* of 1 mm will reduce diameter by 2 mm

Because our workpiece was longer, we had to face and center drill the free end and use a dead or live center in the tailstock to support it. Without such support, the force of the tool on the workpiece would cause it to bend away from the tool, producing a strangely shaped result. There is also the potential that the work could be forced to loosen in the chuck jaws and fly out as a dangerous projectile.

We inserted the workpiece in the 3-jaw chuck and tightened down the jaws until they just start to grip the workpiece. We then rotated the workpiece to ensure that it is seated evenly and to dislodge any chips or grit on the surface that might keep it from seating evenly. You want the workpiece to be as parallel as possible

with the center line of the lathe. We tightened the chuck using each of the three chuck key positions to ensure a tight and even grip.

3.4 ADJUSTING THE TOOL BIT

We chose a tool bit with a slightly rounded tip. This type of tool produces a nice smooth finish. For more aggressive cutting, when we needed to remove a lot of metal, we chose a tool with a sharper tip. The tool was tightly clamped in the toolholder.

We then adjusted the angle of the tool holder so the the tool is approximately perpendicular to the side of the workpiece. Because the front edge of the tool is ground at an angle, the left side of the tip should engage the work, but not the entire front edge of the tool.

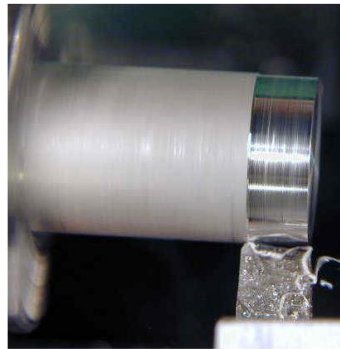


Figure 3.3: Turning process

We turned two steel shaft 20mm and welded them on both sides of our shaft. It is this reduced part that entered the rims through the wheel hub.



Figure 3.4: machined shaft

Facing

Facing is the process of removing metal from the end of a workpiece to produce a flat surface. Most often, the workpiece is cylindrical, but using a 4-jaw chuck you can face rectangular or odd-shaped work to form cubes and other non-cylindrical shapes.

When a lathe cutting tool removes metal it applies considerable tangential (i.e. lateral or sideways) force to the workpiece. To safely perform a facing operation the end of the workpiece must be positioned close to the jaws of the chuck. The workpiece should not extend more than 2-3 times its diameter from the chuck jaws unless a steady rest is used to support the free end.

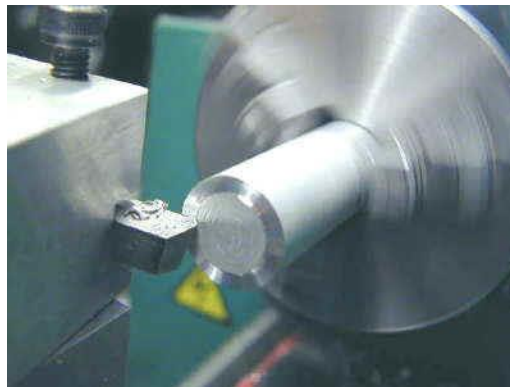


Figure 3.5: Facing process

This was done before turning the shaft.

Centre drilling

Drilling Operations

The alignment between the headstock and tailstock of the lathe enables drilling of holes that are precisely centered in a cylindrical piece of stock. Doing this directly with a drill does not turn out too well hence a lathe machine is very important.

Before drilling we first faced the ends of the workpiece as described above. The next step was to start the drill hole using a center drill - a stiff, stubby drill with a short tip. If you try to drill a hole without first center drilling, the drill will almost certainly wander off center, producing a hole that is oversized and misaligned. That's unwanted!



Figure 3.6: Various sizes of centre drills.

3.5 PREPARING TO DRILL

Before drilling we made sure that the drill chuck is firmly seated in the tailstock. With the chuck arbor loosely inserted in the tailstock bore, we cranked the tailstock bore out about 1/2". We then locked the tailstock to the ways, then thrust the chuck firmly back towards the tailstock to firmly seat the arbor in the Morse taper of the tailstock. (The chuck is removed from the tailstock by cranking the tailstock ram back until the arbor is forced out).

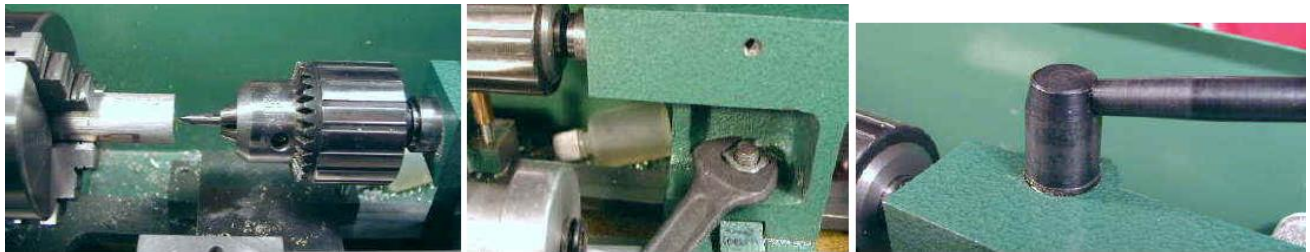


Figure 3.7: Drilling processes

We used a center drill of diameter (10mm), similar to that of the hole that we intend to drill. We then Insert the center drill in the jaws of the tailstock chuck and tighten the chuck until the jaws just start to grip the drill. Since the goal is to make the drill as stiff as possible, we didn't want it to extend very far from the tip of the jaws.



Figure 3.8: Partly drilled hole necessary before turning operations

Later we twisted the drill to seat it and dislodge any metal chips or other crud that might keep the drill from seating properly. Now tighten the chuck. It's good practice to use 2 or 3 of the chuck key holes to ensure even tightening (but all three may be impossible to reach given the tight confines of the 7x10).

Slide the tailstock along the ways until the tip of the center drill is about 1/4" from the end of the workpiece and tighten the tailstock clamp nut. The locking lever for the tailstock ram should be just snug - not enough to impede the movement of the ram, but enough to ensure that the ram is as rigid as possible.



Figure 3.9: Our team performing drilling operations

3.6 FABRICATION OF WASHERS



Figure 3.10: washer

A **washer** is a thin plate (typically disk-shaped, but sometimes square) with a hole (typically in the middle) that is normally used to distribute the load of a threaded fastener, such as a bolt or nut. Other uses are as a spacer, spring (Belleville washer, wave washer), wear pad, preload indicating device, locking device, and to reduce vibration (rubber washer). Washers often have an outer diameter (OD) about twice their inner diameter (ID), but this can vary quite widely.

We preferred to fabricate our washers rather than buy. We fabricated washers to have 12.7 mm internal diameter. We achieved this by

- a. Identifying two thin plates and welding them together.



Figure 3.11: Thin plates for making a washer

- b. Using two diagonals, we identified the centre. This center was the one used in drilling.



Figure 3.12: Diagonals on the two plates to identify centre

- c. Using lathe machine, we drilled a 12.8 mm through hole.
- d. Through facing, we smoothened its surface
- e. We cut off a circular shape through turning.



Fig 3.13: Making a washer with lathe machine

3.7 FABRICATION OF TURBINE BLADE

Savonius blades are a crucial and basic part of a wind turbine. They are mainly made of aluminum, fiber glass or carbon fiber. Strong Plastic pipe was selected because they provide better strength to weight ratio.

The design of the individual blades also affects the overall design of the rotor. Rotor blades take the energy out of the wind; they capture the wind and convert its kinetic energy into rotation of the hub.

We used a one meter pipe.

We cut it into two halves.

We then made two holes in which cables will pass through and fix them into position.



Figure 3.14: Scoops

3.8 FABRICATING LOWER SUPPORT SHAFT



Figure 3.15: Lower Support Shaft

This shaft is mainly for supporting the turbine through rims to the base.

We first made a C-disk. Through boring, we then threaded its internal surface. Also the aim of boring was to help us achieve the intended diameter of 24 mm so that one of our bearings can fit inside. In machining, **boring** is the process of enlarging a hole that has already been drilled (or cast) by means of a single-point cutting tool (or of a boring head containing several such tools), such as in boring a gun barrel or an engine cylinder. Boring is used to achieve greater accuracy of the diameter of a hole, and can be used to cut a tapered hole. Boring can be viewed as the internal-diameter counterpart to turning, which cuts external diameters. Boring, also called **internal turning**.



Figure 3.16: Boring tool

3.9 MODIFICATION OF HUB

A hub is the center part of a bicycle wheel. It consists of an axle, bearings and a hub shell. The hub shell typically has two machined metal flanges to which spokes can be attached. We removed the central axle and instead used the shaft we had machined. It's the shaft that acted as the axle.

3.10 MODIFICATION OF BEARINGS

The bearings allow the hub shell (and the rest of the wheel parts) to rotate freely about the axle. We used two bearings each of 12mm diameter.

We welded the inner part to our shaft so that as the shaft rotates, the inner part of the bearing also rotates while the outside remains rigid.



Figure 3.17: Bearing

3.11 HUB SHELL AND FLANGES

The hub shell is the part of the hub to which the spokes (or disc structure) attach. The hub shell of a spoked wheel generally has two flanges extending radially outward from the axle. Each flange has holes or slots to which spokes are affixed.

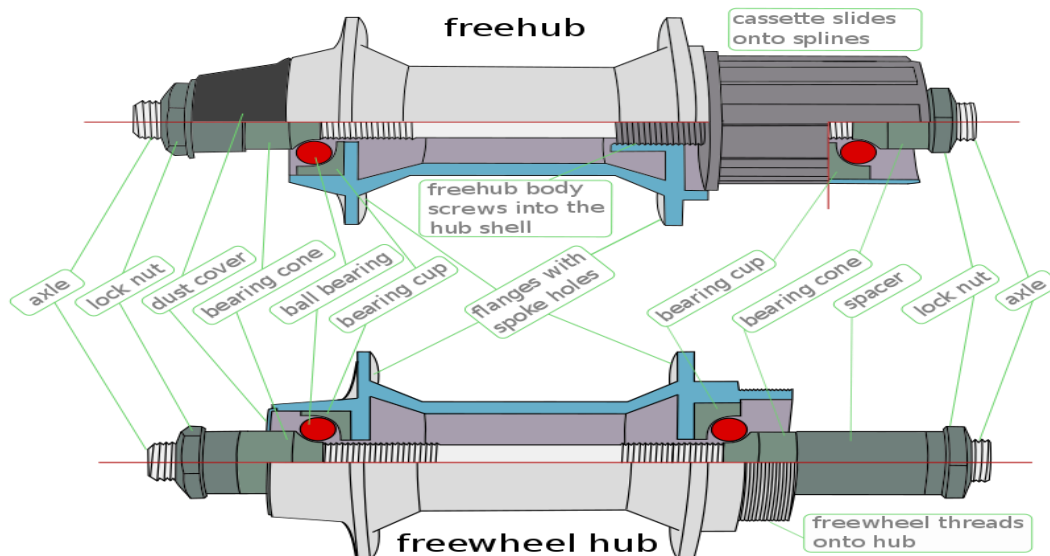


Figure 318: Hub Shell

3.12 RIM



Figure 3.19: RIM

The **rim** is the "outer edge of a wheel, holding the tire". It makes up the outer circular design of the wheel on which the inside edge of the tire is mounted on vehicles such as automobiles. Two rims were used. Blades/scopes was attached to the rim.

We used 26 by 1.25 inch diameter.

3.13 SPOKES

The rim was connected to the hub by several spokes under tension.

The spokes used was stainless steel. Stainless steel spokes are favored by most manufacturers and riders for their durability, stiffness, damage tolerance, and ease of maintenance. A total of 32 spokes were used.

Operational Procedure

- We used Sun Singletrack rims and Kidride hubs.



Figure 3.20: kidride hub

- The first spoke was then popped in from the outside of the hub and laced through the rim hole on the right nearest the valve hole.
- The next spoke was popped into the hub from the same side but one hole along.
- 4 holes from the first spoke was counted and another spoke put into the 4th hole.
- After holes on one side of the hub are full, twist the hub. Twist in the right direction the spoke nearest your valve hole runs almost 90 degrees from the rim.
- The process was repeated on the other side of the hub.
- The shaft was then welded between the two rims.

- Another smaller shaft was welded on the lower rim.
- Support for the rims was then welded and the blade fitted into position.
- By use of drills, 1mm holes were made on the blades.

CONNECTING THE ALTERNATOR

An **alternator** is an electrical generator that converts mechanical energy to electrical energy in the form of alternating current. For reasons of cost and simplicity, most alternators use a rotating magnetic field with a stationary armature. Occasionally, a linear alternator or a rotating armature with a stationary magnetic field is used. In principle, any AC electrical generator can be called an alternator, but usually the term refers to small rotating machines driven by automotive and other internal combustion engines.



Figure 3.21: alternator

Connect the device to a 12-volt battery to provide stability in the system. Connect the largest terminal of the alternator to the positive terminal of the battery; the negative terminal of the battery connects to the frame (housing) of the alternator. Two or three smaller terminals will remain depending on the design.

With the two-terminal design it usually means that a regulator is built in, and you need to attach one terminal to battery positive and the other small terminal, through some type of switch, to a positive voltage source. This turn-on circuit energizes the regulator and will control the alternator output voltage to a maximum of approximately 14 to 14.7 volts.

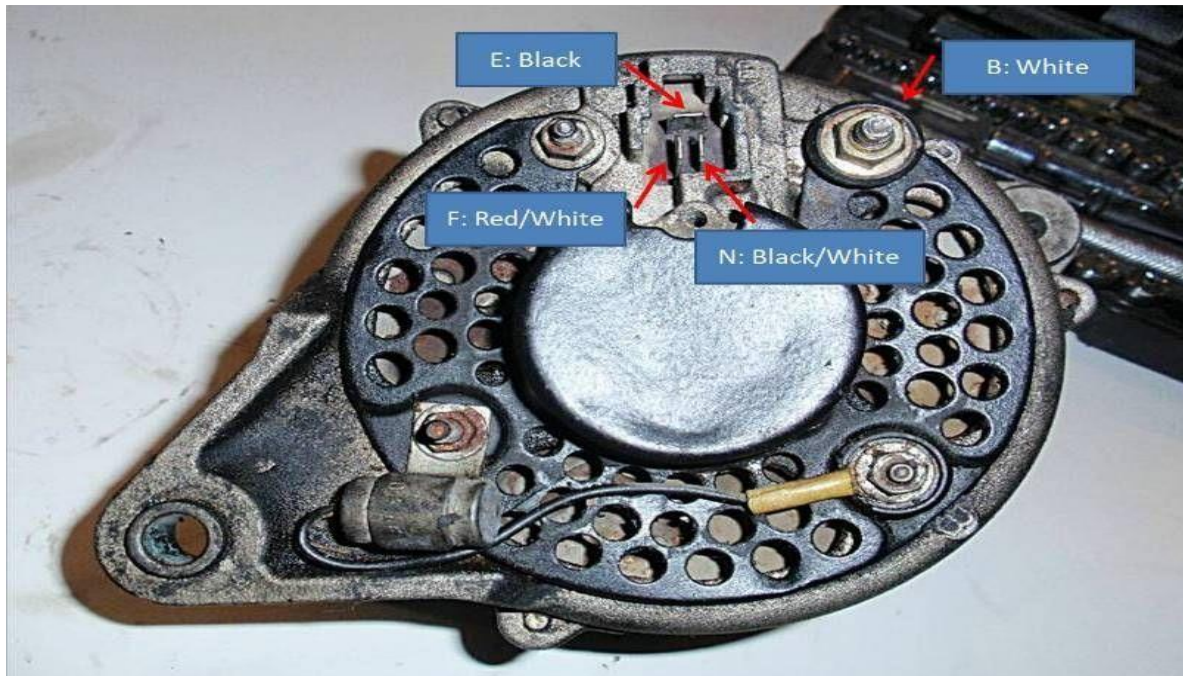


Figure 3.22: Bottom of alternator

With a three-terminal design it usually means you have to mount an external regulator and run the corresponding wiring from the alternator to the regulator and the battery positive. Turn on positive voltage, and then go to the regulator instead of the alternator

Connect the large red terminal on the alternator to the positive terminal of the battery. It is the one written “**B**”. Connect the negative wire of the alternator to the negative side of the battery. Connect the two earth cables together. As the gas engine runs, it will recharge the battery.

Connect the red positive terminal of the power inverter to the positive terminal of the battery. Finally, connect the black negative terminal of the wire to the power inverter. The size of the power inverter is dependent on how much equipment you plan on running off it. Using a very large power inverter may require additional batteries. Small inverters typically have electrical outlets to plug appliances into. Large inverters can be wired to a power panel.

3.14 HOW THE ALTERNATOR WORKS

The alternator consists of a stator - a stationary set of wire coil windings, inside which a rotor revolves.

The rotor is an electromagnet supplied with a small amount of electricity through carbon or copper-carbon brushes (contacts) touching two revolving metal slip rings on its shaft.

The rotation of the electromagnet inside the stator coils generates much more electricity inside these coils.

The electricity is alternating current - its direction of flow changes back and forth every time the rotor turns. It has to be rectified - turned into a one-way flow, or direct current.

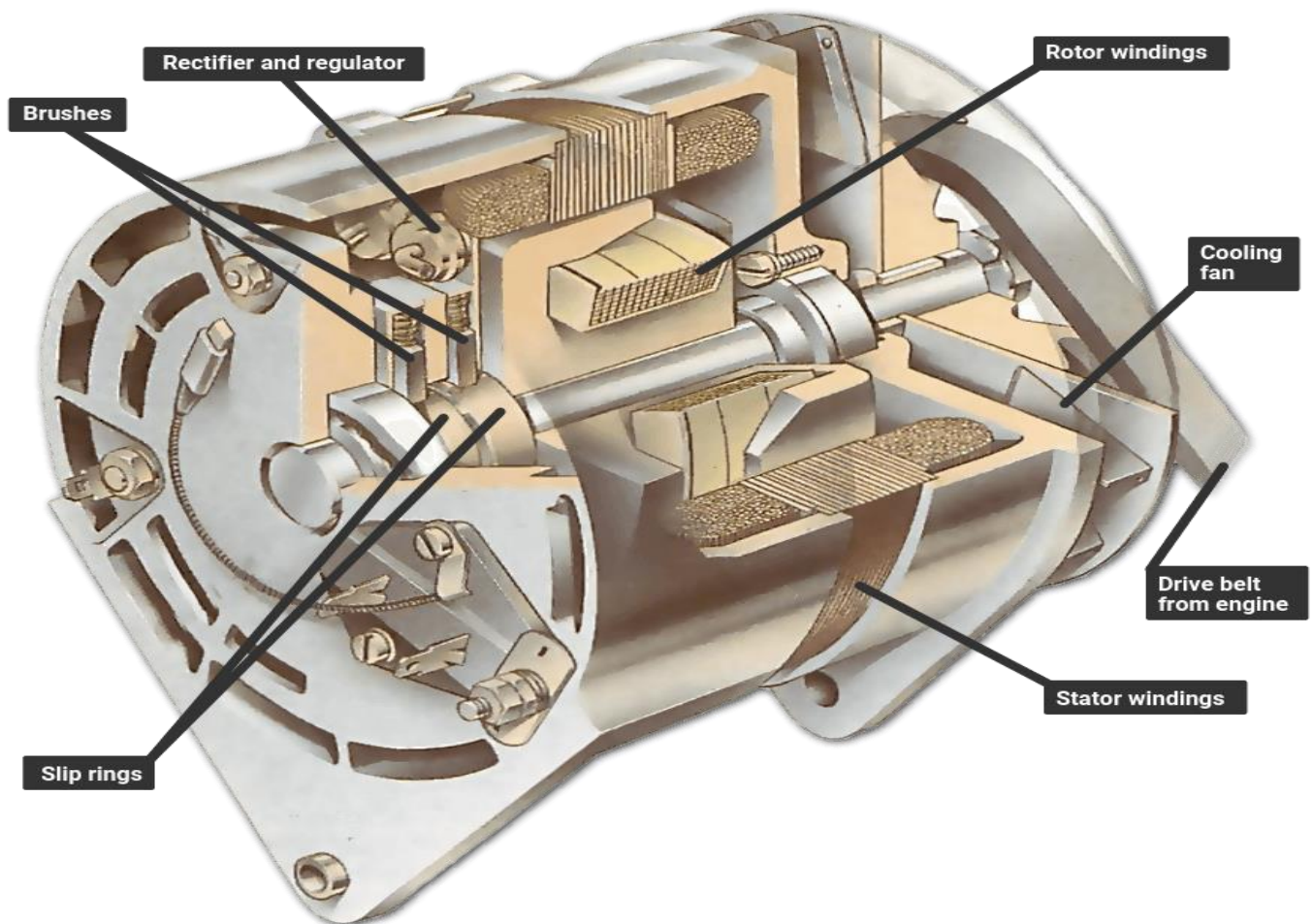


Fig 3.23: Inside an alternator

A dynamo gives direct current but is less efficient, particularly at low engine speeds, and weighs more than an alternator.

Moving a magnet past a closed loop of wire makes an electric current flow in the wire. Imagine a loop of wire with a magnet inside it.

The north pole of the magnet passes the top of the loop as the South Pole passes the bottom of it. Both passes make current flow in one direction round the loop.

The poles move away, and current stops flowing until the South Pole reaches the top and the North Pole the bottom.

This makes current flow again, but in the opposite direction.

A car alternator uses an electromagnet in order to boost output of electric current.

3.15 REGULATING THE CURRENT TO THE BATTERY

The current from an alternator is rectified into direct current by a set of diodes that allow current to flow through them in one direction only.

To charge the battery the voltage supplied to it must not be too low or too high.

The alternator has a transistor-operated control device that regulates the voltage by supplying more or less current - as required - to the electromagnet.

The rectifier and regulator are usually inside the alternator housing, but on some alternators they are outside, mounted on the alternator body.

BATTERY

A conventional starter battery consists of 6 cells connected in series, each with a nominal voltage of 2 V, which results in a voltage of exactly 12.72 V when the battery is fully charged. The capacity and the cold start capability of the battery results from the number of plates per cell.

Rule of thumb: The more plates which a cell contains, and therefore form a larger surface, the larger the cold start power (CCA) which the battery can deliver. However, if the space in the cell is used for fewer, but thicker plates, the cycle stability is increased. This means that the battery is designed for a higher charge throughput (continual charging and discharging process).

The cells are contained in a casing which is made from acid-resistant plastic (polypropylene). In a conventional SLI battery, this is closed with a cover with a labyrinth system which prevents the battery fluid from escaping and separates the liquid from gas.

Early batteries had screw plugs which enabled them to be topped up with distilled water. Modern batteries are completely maintenance-free. Water does not need to be, and must not be topped up. Although AGM batteries still have “one-way plugs”, these must not be opened under any circumstances.

3.16 CAR BATTERY FUNCTION: CHEMICAL ENERGY BECOMES ELECTRICAL ENERGY

A car battery stores energy in chemical form and converts it into electrical energy. In this electro-chemical process, four materials react with each other:

- Hydrogen (H)
 - Oxygen (O₂)
 - Lead (Pb)
 - Sulfur (S)
1. A typical SLI battery has six cells. Each cell has two plates, or grids: one is made of lead, the other of lead dioxide. Each cell is able to produce about 2-volts of energy. In most car batteries you have six cells, and therefore a 12-volt battery.
 2. The plates are submerged in sulphuric acid that triggers a reaction between the two plates. In scientific terms, the acid acts as a catalyst.
 3. This acid will trigger a reaction on the lead dioxide plate, causing the plate to produce two things: ions and lead sulphate.
 4. The ions produced by the lead dioxide plate react to the adjacent plate to produce hydrogen and lead sulphate.

5. The result is a chemical reaction that produces electrons. The electrons race around the plates and generate electricity. The electricity flows out of the battery terminals to start your engine, turn on your headlights, and play the radio.
6. This chemical reaction is entirely reversible, which is why you can jumpstart your battery and continue to charge it throughout the duration of its life. By applying current to the battery at just the right voltage, lead and lead dioxide will form on the plates and you can reuse your battery, over and over again!

We used a 12 volt DC battery.



Figure 3.24: Battery

Boling the Alternator

Bolt the alternator to the deck, and attach a pulley on the output shaft of the gas motor to drive the alternator through a belt.

Connect the alternator to the turbine using a V-belt. As the turbine runs, a pulley on the gas engine shaft will turn the pulley on the alternator which allows the alternator to produce power.

Summary of activities

1. Joining spokes to the wheel hub

2. Connecting the spokes to the rim
3. Aligning the spokes with the wheels
4. Turning 20 mm on both sides of two solid steel shafts.
5. Welding the two solid shafts on our main hollow shaft
6. Fabricating our washers.
7. Inserting the washers on opposite sides of our steel shaft.
8. Inserting the two rims.
9. Inserting two bearings on both sides.
10. Attaching base.
11. Attaching alternator with rim through v-belt.
12. Attaching Arduino speed sensor.

FINAL APPEARANCE OF OUR FABRICATED VAWT



Figure 3.25: final part

CHAPTER 4

RESULTS

4.1 THEORY

Wind turbines work by converting the kinetic energy in the wind first into rotational kinetic energy in the turbine and then electrical energy that can be supplied, via the national grid. The energy available for conversion mainly depends on

1. the wind speed
2. Swept area of the turbine.
3. Air density

4.2 MATHEMATICAL MODEL

The following table shows the definition of various variables used in this model:

E = Kinetic Energy (J)	ρ = Density (kg/m ³)
m = Mass (kg)	A = Swept Area (m ²)
v = Wind Speed (m/s)	C_p = Power Coefficient
P = Power (W)	r = Radius (m)
Dm/dt = Mass flow rate (kg/s)	x = distance (m)
DE/dt = Energy Flow	t = time (s)

4.2.1 Rate (J/s)

Under constant acceleration, the kinetic energy of an object having mass m and velocity v is equal to the work done W in displacing that object from rest to a distance s under a force F , i.e.:

$$E = W = Fs$$

According to Newton's Law, we have:

$$F = ma$$

Hence,

$$E = mas \dots (1)$$

Using the third equation of motion:

$$v^2 = u^2 + 2as$$

we get:

$$a = \frac{(v^2 - u^2)}{2s}$$

Since the initial velocity of the object is zero, i.e. $u=0$, we get:

$$a = \frac{v^2}{2s}$$

Substituting it in equation (1), we get that the kinetic energy of a mass in motions is:

$$E = \frac{1}{2}mv^2 \dots (2)$$

The power in the wind is given by the rate of change of energy:

$$P = \frac{dE}{dt} = \frac{1}{2}v^2 \frac{dm}{dt} \dots (3)$$

As mass flow rate is given by

$$\frac{dm}{dt} = \rho A \frac{dx}{dt}$$

we get:

$$\frac{dm}{dt} = \rho A v$$

Hence, from equation (3), the power can be defined as:

$$P = \frac{1}{2} \rho A v^3 \dots (4)$$

A German physicist Albert Betz concluded in 1919 that no wind turbine can convert more than 16/27 (59.3%) of the kinetic energy of the wind into mechanical energy turning a rotor. To this day, this is known as the **Betz Limit** or **Betz' Law**. The theoretical maximum **power efficiency** of *any* design of wind turbine is

0.59 (i.e. no more than 59% of the energy carried by the wind can be extracted by a wind turbine). This is called the “power coefficient” and is defined as:

$$C_{p_{max}} = 0.59$$

Also, wind turbines cannot operate at this maximum limit. The C_p value is unique to each turbine type and is a function of wind speed that the turbine is operating in. Once we incorporate various engineering requirements of a wind turbine - strength and durability in particular - the real world limit is well below the *Betz Limit* with values of 0.35-0.45 common even in the best designed wind turbines. By the time we take into account the other factors in a complete wind turbine system - e.g. the gearbox, bearings, generator and so on - only 10-30% of the power of the wind is ever actually converted into usable electricity. Hence, the power coefficient needs to be factored in equation (4) and the extractable power from the wind is given by:

$$P_{avail} = \frac{1}{2} \rho A v^3 C_p \dots (5)$$

The swept area of the turbine can be calculated from the length of the turbine blades using the equation for the area of a circle:

$$A = \pi r^2 \dots (6)$$

4.3 CALCULATIONS WITH GIVEN DATA

We are given the following data:

Blade length, $l = 1$ m

Wind speed, $v = 12$ m/sec

Air density, $\rho = 1.23$ kg/m³

Power Coefficient, $C_p = 0.4$

Inserting the value for blade length as the radius of the swept area into equation (6) we have:

$$l = r = 1\text{m}$$

$$A = \pi r^2$$

$$= \pi \times 1$$

$$= 3.142857\text{m}^2$$

We can then calculate the power converted from the wind into rotational energy in the turbine using equation (7):

$$P_{avail} = \frac{1}{2} \rho A v^3 C_p$$

$$= 0.5 \times 1.23 \times 3.142857 \times 12^3 \times 0.4$$

=1336W

=1.3KW

ACTUAL/ OBSERVED VALUES

Using our Arduino infrared sensor, we were able to measure the revolutions per minute on wind turbine.

Using a tachometer, we were able to measure the revolutions produced by alternator per second

For battery charging to occur, the alternator's voltage must exceed the battery's voltage.

Alternator may not generate sufficient charging voltage until alternator speed is greater than about 2000 RPM. Typically, alternators have their full output rated at 6000 RPM but can continue to spin up to 12,000 RPM or more without any additional increase in output. The speed of an alternator is different for different type of car. The speed of an alternator depends on the speed of the engine. For a racing car, the ratio speed between engine and alternator is usually 1:1. For a drag car, the ratio is usually 1:2. And for street use, the ratio is usually 1:3.

When voltage causes current to flow, energy is converted. This is described as power.

The unit of power is the watt. As with Ohm's law, any one value can be calculated if the other two are known.

Power = Voltage X Current (1)

$P = VI$ or $I = P/V$ or $V = P/I$

Our VAWT charging system has been represented in Figure 10 as three blocks, i.e. the alternator, battery, and loads. When turbine is not running, the alternator voltage is less than the battery voltage so current flows from the battery to the household loads and the alternator diodes prevent current flowing into the alternator. When engine is running, the alternator output is greater than the battery voltage, so current flows from the alternator to the vehicle loads as well as the battery. This implies that alternator output voltage must always be above the battery voltage during operation of the engine. However, the actual Voltage used is critical and depends on a number of factors.

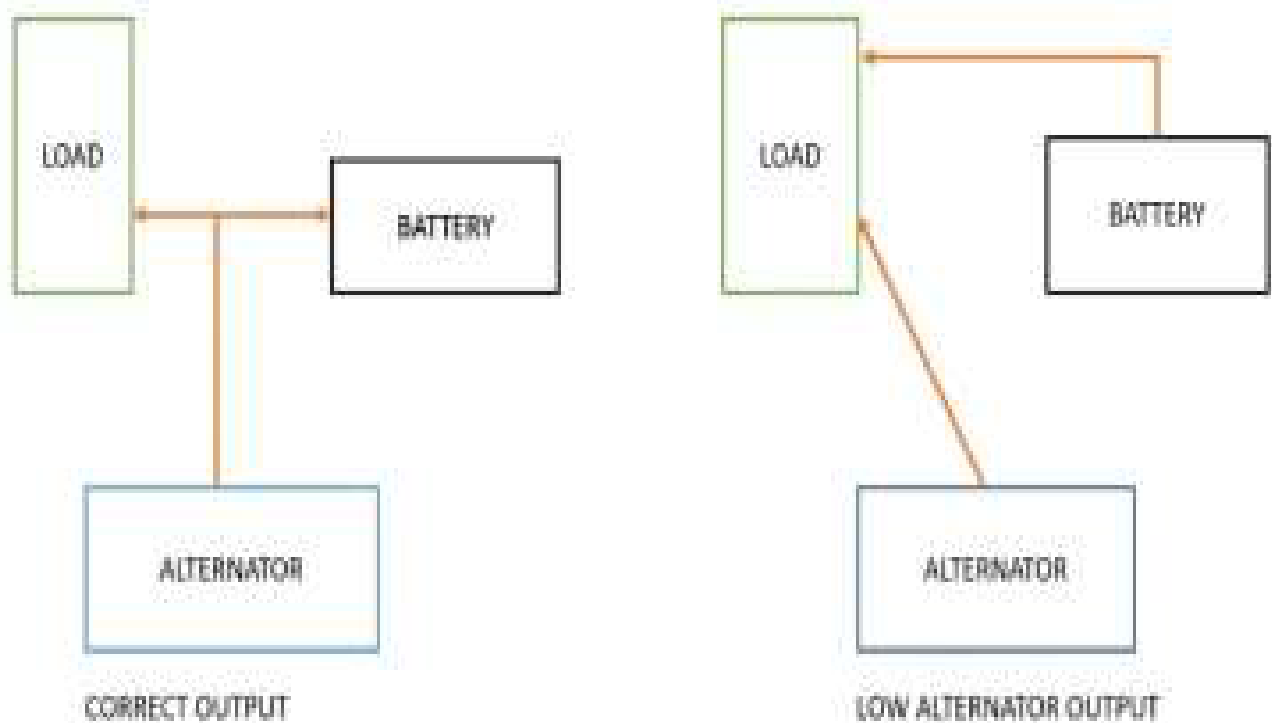


Figure 4.1: Block diagram representing our VAWT charging system.

The relationship between alternator speed and alternator current is shown in Figure 15. The current output of the alternator is speed-dependent.

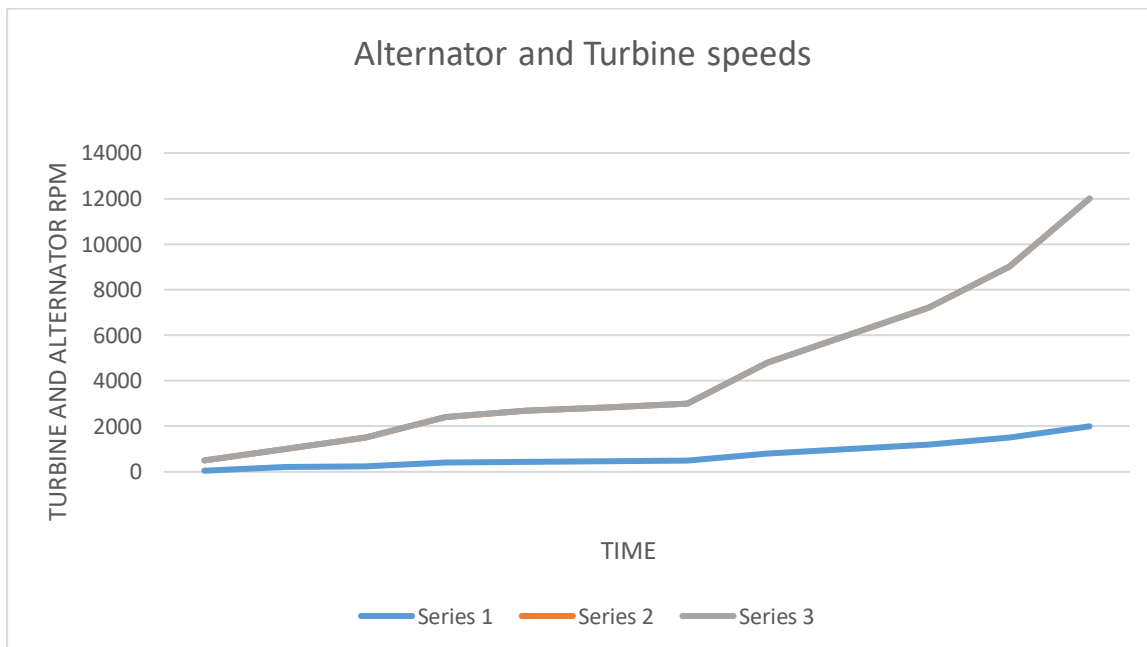
The greater the speed, the greater the output. The rated current is output at an alternator speed of 6000 RPM.

Using our Arduino infrared sensor, we were able to measure the revolutions per minute on wind turbine.

Using a tachometer, we were able to know the revolutions produced by alternator per second.

Table 4.1: Speed difference between engine and alternator.

Engine speed (RPM)	Alternator speed (RPM)
50	450
200	800
250	1250
400	2000
450	2250
470	2350
500	2500
800	4000
1000	5000
1200	6000
1500	7500
2000	10000



Series 1 is Alternator speed

Series 2 is Turbine speed

Table 2. Battery Voltages

RPM	VOLTAGE	CURRENT
500	4	65
1000	6	100
5000	7.3	105
8000	8	120

But

Power = Voltage X Current (1)

$P = VI$

Therefore using multiplication, the various powers produced are

VOLTAGE (V)	CURRENT (I)	POWER (W)
4	35	140
6	50	300
6.5	55	358
6.7	57	382

EFFICIENCY

Efficiency is the state or quality of being efficient.

To get the efficiency, both the actual power generated and theoretical power expected were analyzed

$(\text{ACTUAL POWER OUTPUT}) / (\text{THEORITICAL POWER OUTPUT}) * 100$

Actual power was **382 Watts**

Theoritical power expected to be produced was **1300 watts**

Therefore

$$(382)/(1300)*100$$

$$=30\%$$

CHAPTER 5

CONCLUSION AND RECOMMENDATION

The efficiency of the fabricated wind turbine was found to be around 30% with variation at different velocities of the wind. This is a very good efficiency compared to horizontal axis wind turbine that has an efficiency of 35%.

The current VAWTs that are already in the market with different designs have an efficiency of 17%.

Though the efficiency seems quite low but it can be seen as usable power generated from nothing. One of the reasons for the low efficiency is that we had much dead weight. These turbines have a great advantage over Horizontal axis wind turbine that they can work on low height thus these turbines can be installed on individual houses for their particular use. The power generated from the turbine can be stored or used to charge a storage device and then the storage device can be used as a continuous power source.

Having said that there are some limitation of these turbines VAWT blades are rarely at an optimal angle to the wind or in clean air, so they can never be as efficient as a tripled HAWT and won't generate more electricity, HAWTs almost never collapse due to lateral stress, and

VAWTs typically have very asymmetrical front and rear stresses on their bearings, and to generate the same electricity, VAWTs would have to be as tall as HAWTs, so visual impact will be virtually identical.

Yet with more and more engineers and researchers working on the design and development of Vertical axis wind turbine the efficiency can be increased and the usability of these turbines can be increased.

RECOMMENDATION

1. The VAWT should be lifted high enough to maximize wind speed.
2. Its blades should be wide to capture good wind speed.
3. Dead weight should be reduced.
4. A high power alternator should be used to provide more power.

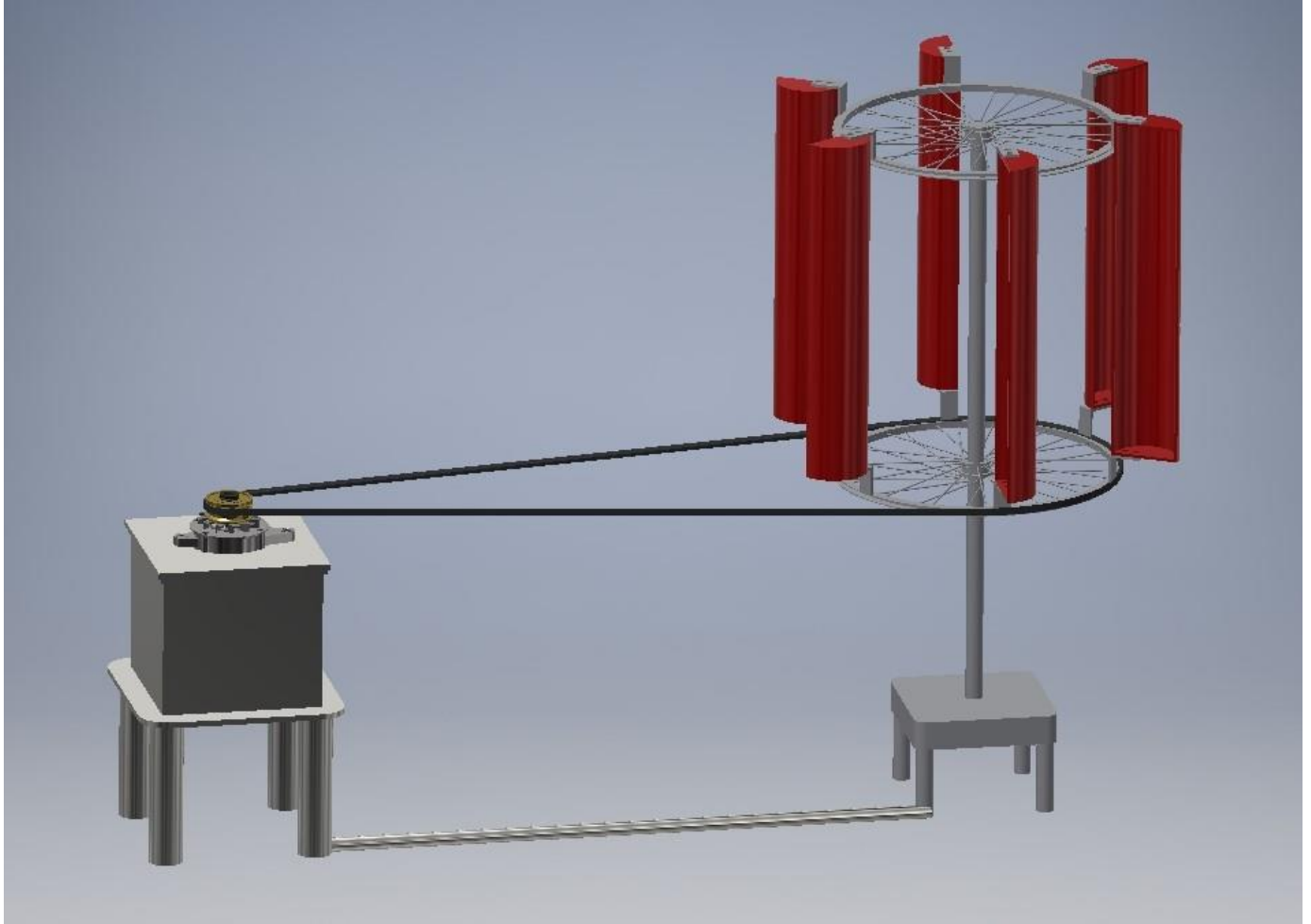
PROBLEMS ENCOUNTERED

1. Wind speed in Moi was very low

2. There was much dead weight in our system i.e. the central shaft which is heavy was also rotating thus reducing the speed.
3. All compressors in Moi University failed thus it was difficult to make a wind tunnel.
4. Inadequate funds to purchase all equipment such as battery

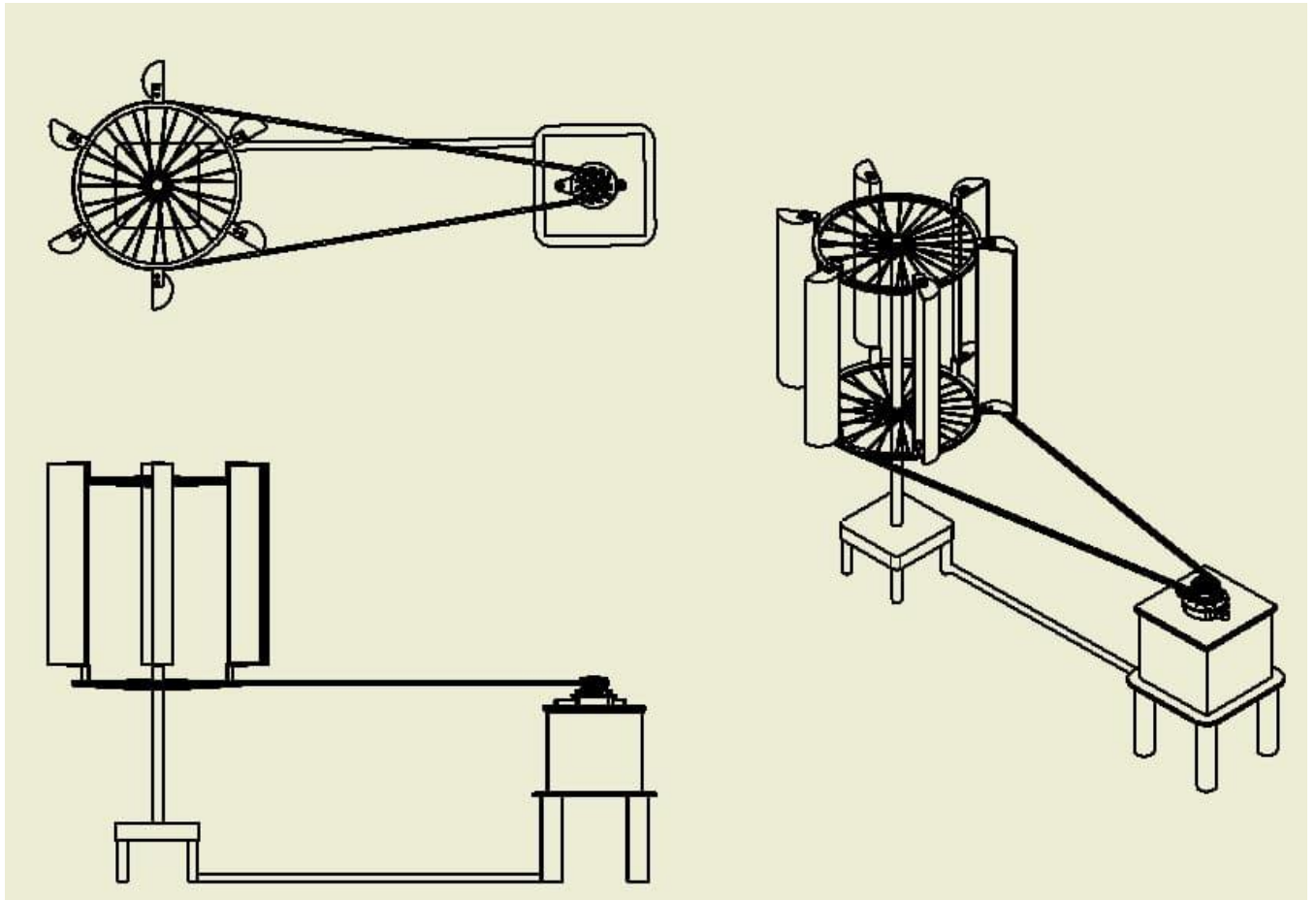
APPENDICES

3D DRAWING OF OUR VAWT SYSTEM



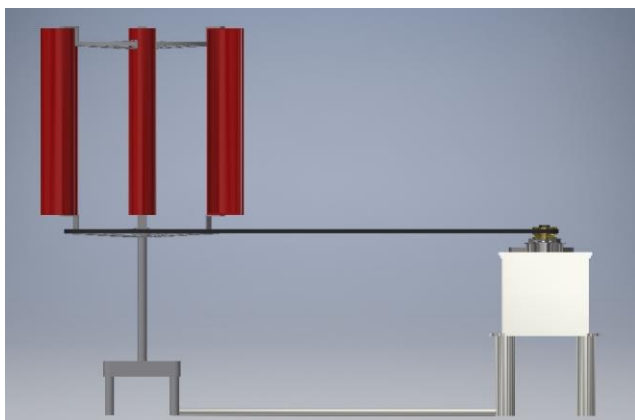
APPENDIX 1

2D DRAWING



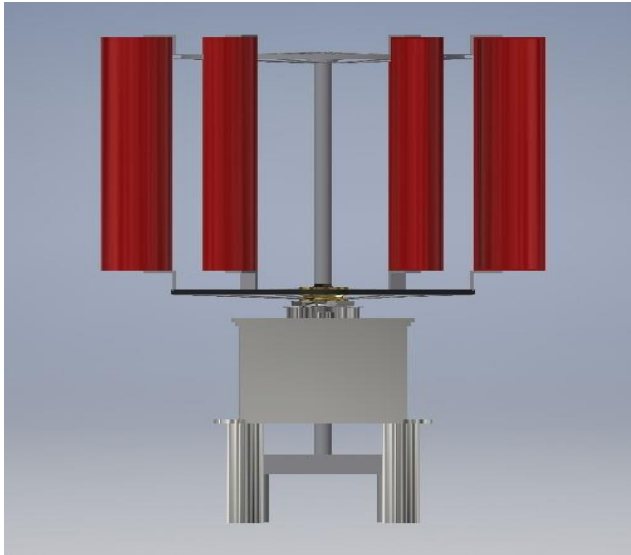
APPENDIX 2

FRONT VIEW



APPENDIX 3

SIDE VIEW



APPENDIX 4

REFERENCES.

1. Hau, E. *Wind Turbines, Fundamentals, Technologies, Application, Economics*, 2nd ed.; Springer: Berlin, Germany, 2006.
2. Dominy, R.; Lunt, P.; Bickerdyke, A.; Dominy, J. Self-starting capability of a darrieus turbine. *Proc. Inst. Mech. Eng. Part A J. Power Energy* **2007**, *221*, 111–120.
3. Holdsworth, B. *Green Light for Unique NOVA Offshore Wind Turbine*, 2009. Available online: <http://www.reinforcedplastics.com> (accessed on 8 May 2012).
4. Gasch, R.; Tvele, J. *Wind Power Plants*; Solarpraxis: Berlin, Germany, 2002.
5. Gorban, A.N.; Gorlov, A.M.; Silantyev, V.M. Limits of the turbine efficiency for free fluid flow. *J. Energy Resour. Technol. Trans. ASME* **2001**, *123*, 311–317.
6. Burton, T. *Wind Energy Handbook*; John Wiley & Sons Ltd.: Chichester, UK, 2011.
7. Hull, D.G. *Fundamentals of Airplane Flight Mechanics*; Springer: Berlin, Germany, 2007.